

Bisphosphonate-associated osteonecrosis of the jaw: report of a task force of the American Society for Bone and Mineral Research.

ONJ has been increasingly suspected to be a potential complication of bisphosphonate therapy in recent years. Thus, the ASBMR leadership appointed a multidisciplinary task force to address key questions related to case definition, epidemiology, risk factors, diagnostic imaging, clinical management, and future areas for research related to the disorder. This report summarizes the findings and recommendations of the task force.

INTRODUCTION: The increasing recognition that use of bisphosphonates may be associated with osteonecrosis of the jaw (ONJ) led the leadership of the American Society for Bone and Mineral Research (ASBMR) to appoint a task force to address a number of key questions related to this disorder.

MATERIALS AND METHODS: A multidisciplinary expert group reviewed all pertinent published data on bisphosphonate-associated ONJ. Food and Drug Administration drug adverse event reports were also reviewed.

RESULTS AND CONCLUSIONS: A case definition was developed so that subsequent studies could report on the same condition. The task force defined ONJ as the presence of exposed bone in the maxillofacial region that did not heal within 8 wk after identification by a health care provider. Based on review of both published and unpublished data, the risk of ONJ associated with oral bisphosphonate therapy for osteoporosis seems to be low, estimated between 1 in 10,000 and <1 in 100,000 patient-treatment years. However, the task force recognized that information on incidence of ONJ is rapidly evolving and that the true incidence may be higher. The risk of ONJ in patients with cancer treated with high doses of intravenous bisphosphonates is clearly higher, in the range of 1-10 per 100 patients (depending on duration of therapy). In the future, improved diagnostic imaging modalities, such as optical coherence tomography or MRI combined with contrast agents and the manipulation of image planes, may identify patients at preclinical or early stages of the disease. Management is largely supportive. A research agenda aimed at filling the considerable gaps in knowledge regarding this disorder was also outlined.